Jiawen (Jarvan) Wang

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PROFILE

Master of Information Technology student at the University of Technology Sydney seeking AI, data analytics, machine learning, or related technology internships. Hands-on Python experience across computer vision, NLP, and healthcare prediction, including data cleaning, feature engineering, modeling, visualization, cross-validation, and metric-based evaluation.

CORE STRENGTHS

- Builds end-to-end data workflows from raw data preparation to feature engineering, analysis, modeling, and error review.
- Evaluates models with Accuracy, F1-score, ROC-AUC, confusion matrices, validation curves, and cross-validation.
- Uses PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, and Jupyter notebooks for reproducible analysis and experiments.

EDUCATION

University of Technology Sydney 2025-Present
Master of Information Technology | Coursework: Database, Data Analytics, Advanced Data Analytics Algorithms, Project Management, Software Development

Huazhong Agricultural University 2019-2023
Bachelor of Biotechnology | Coursework: Microbiology, Cell Biology, Genetics, Biochemistry, Genetic Engineering, Pharmacology

PROJECTS

Deep Learning for Image Classification and Object Detection
PyTorch, ResNet18, Faster R-CNN, SSD300

- Built PyTorch pipelines for image classification and object detection with **ResNet18, Faster R-CNN, and SSD300**.
- Compared baseline and customized ResNet18 architectures using training/validation loss and accuracy curves to assess convergence and overfitting.
- Visualized dataset samples, annotations, predictions, and bounding boxes to support model debugging and error analysis.

Handwritten Mathematical Symbol Recognition
Python, Scikit-learn, OpenCV, HOG, LBP

- Engineered HOG, LBP, and raw-pixel feature representations for a multi-class symbol recognition workflow.
- Trained SVM and ANN classifiers; compared confusion matrices and accuracy metrics to identify misclassification patterns.

Cardiovascular Disease Prediction
Python, Scikit-learn, Pandas, NumPy

- Cleaned and engineered features from **56,000+ medical records**, including missing-value handling and structured preprocessing.
- Built Logistic Regression, Random Forest, and KNN models; compared Accuracy, F1-score, and ROC-AUC to select the strongest model.

SMS Spam Classification
Python, Scikit-learn, NLP, TF-IDF

- Processed **5,574 SMS messages** using TF-IDF vectorization for binary text classification.
- Implemented Naive Bayes and Perceptron models with Stratified K-Fold cross-validation and F1-score analysis.

TECHNICAL SKILLS

AI / Data / ML: Scikit-learn, PyTorch, supervised learning, classification, model evaluation, feature engineering, cross-validation, error analysis

Deep Learning & Computer Vision: CNN, ResNet18, Faster R-CNN, SSD300, OpenCV, image classification, object detection

Programming, Tools & Languages: Python, SQL, Pandas, NumPy, Matplotlib, Seaborn, TF-IDF, Jupyter Notebook, Git, GitHub, VS Code | English, Mandarin Chinese